ICT703 – Module 2 Workshop

Learning Objectives

In this Workshop you will:

1. *Design and write a program*
2. *Practice your problem-solving skills in pseudocode*
3. *Practice your python*

1. Income Tax Calculator

This case study requires you to develop a program that calculates income tax. Each year, nearly everyone with an income faces the unpleasant task of computing his or her income tax return. We will go through the software development phases to create the program. We start with the customer request phase

Request

The customer requests a program that computes a person's income tax

Analysis

Analysis often requires the programmer to learn some things about the problem domain, in this case, the relevant tax law. For the sake of simplicity, let's assume the following tax laws:

- All taxpayers are charged a flat tax rate of 20%.
- All taxpayers are allowed a $10,000 standard deduction
- For each dependent, a taxpayer is allowed an additional $3,000 deduction.
- Gross income must be entered to the nearest cent.
- The income tax is expressed as a decimal number.

Another part of analysis determines what information the user will have to provide. In this case, the user inputs are gross income and number of dependents. The program calculates the income tax based on the inputs and the tax law and then displays the income tax. Below is the proposed output. Characters in italics indicate user inputs. The program prints the rest.

Enter the gross income: *150000.00*

Enter the number of dependents: *3*

The income tax is $26200.0

Design

During analysis, we specify what a program is going to do. In the next phase, design, we describe how the program is going to do it. This usually involves writing an algorithm. In Module 1, we showed how to write algorithms in ordinary English. In fact, algorithms are more often written in a somewhat stylized version of English called pseudocode. Write the pseudocode for our income tax program.

Although there are no precise rules governing the syntax of pseudocode, in your pseudocode you should strive to describe the essential elements of the program in a clear and concise manner.

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Implementation (Coding)

Once you have written the preceding pseudocode, write the Python program.

Testing

Our income tax program can run from VS Code. If there are no syntax errors, we will be able to enter a set of inputs and view the results. However, a single run without syntax errors and with correct outputs provides just a slight indication of a program's correctness. Using the sample data in the table below, test your program.

If there is a logic error in the code, it will almost certainly be caught using these data. Note that the negative outputs are not considered errors.

Number of Dependents	Gross Income	Expected Tax
0	10000	0
1	10000	–600
2	10000	–1200
0	20000	2000
1	20000	1400
2	20000	800